How Chatbots Work

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: neural network next word tokens true word

1. Finish the sentence. One word each.

Peanut butter and _____ Once upon a _____ I am going to the _____

2. Count what comes next. The training text:

> I like rice. I like soccer. I like rice and beans. My friend likes soccer. My friend likes rice.

Finish the table. Count how many times each word comes next. ("end" means the end of a sentence.)

After this word	These words came next (how many times)	Dice
I	like (3)	always "like"
like		1 to 4: rice, 5 or 6: soccer
likes		1 to 3: soccer, 4 to 6: rice
rice		1 to 4: end, 5 or 6: and
soccer		always end
and		always "beans"
beans		always end
My		always "friend"
friend		always "likes"

The dice column gives more common words more numbers. After "like": rice _____ %, soccer _____ %

3. Tiny chatbot, real chatbot.

Both chatbots predict the _____. Neither one checks if a sentence is _____.

A real chatbot reads _____: words and pieces of words.

A real chatbot is a _____. Its output layer has one neuron for every _____.

4. The dice chatbot. Start with **I** or **My**. Roll a die when there's a choice.

Sentence 1: _____

Sentence 2: _____

Sentence 3: _____

Did you write a sentence that is **not** in the training text? Which one? _____

5. Talk about it. Did the tiny chatbot understand "My friend likes rice and beans"?

The tiny chatbot **did** / **did not** understand, because _____

_____

_____

Go Further (Day 2). The word-counting chatbot on screen.

With "no chance," is the sentence the same every time? _____

With "a lot of chance," the bars get _____. Rare words win _____.

A strange sentence it wrote: _____

ELL A sentence it wrote in my home language: _____